Hamming Distance Analysis

Set of Bitstrings: $S = \{001010, 110010, 110000, 010001\}$

Introduction

The **Hamming distance** between two bitstrings of the same length is defined as the number of positions in which the corresponding bits differ.

Given the set:

$$S = \{001010, 110010, 110000, 010001\}$$

we want to compute the **minimal** and **maximal** Hamming distances between any two elements.

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Step 1: List all pairs

The set contains 4 bitstrings, so there are $\binom{4}{2} = 6$ unique pairs:

1. 001010 vs 110010,
2. 001010 vs 110000,
3. 001010 vs 010001,
4. 110010 vs 110000,
5. 110010 vs 010001,
6. 110000 vs 010001.

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Step 2: Compute Hamming distances step-by-step

We compare each pair bit by bit using a tabular format.

Pair 1: 001010 vs 110010

Position	001010	110010	Differ?
1	0	1	Yes
2	0	1	Yes
3	1	0	Yes
4	0	0	No
5	1	1	No
6	0	0	No

$$\text{Hamming distance} = 3$$

Pair 2: 001010 vs 110000

Position	001010	110000	Differ?
1	0	1	Yes
2	0	1	Yes
3	1	0	Yes
4	0	0	No
5	1	0	Yes
6	0	0	No

$$\text{Hamming distance} = 4$$

Pair 3: 001010 vs 010001

Position	001010	010001	Differ?
1	0	0	No
2	0	1	Yes
3	1	0	Yes
4	0	0	No
5	1	0	Yes
6	0	1	Yes

$$\text{Hamming distance} = 4$$

Pair 4: 110010 vs 110000

Position	110010	110000	Differ?
1	1	1	No
2	1	1	No
3	0	0	No
4	0	0	No
5	1	0	Yes
6	0	0	No

$$\text{Hamming distance} = 1$$

Pair 5: 110010 vs 010001

Position	110010	010001	Differ?
1	1	0	Yes
2	1	1	No
3	0	0	No
4	0	0	No
5	1	0	Yes
6	0	1	Yes

$$\text{Hamming distance} = 3$$

Pair 6: 110000 vs 010001

Position	110000	010001	Differ?
1	1	0	Yes
2	1	1	No
3	0	0	No
4	0	0	No
5	0	0	No
6	0	1	Yes

$$\text{Hamming distance} = 2$$

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Step 3: Minimal and Maximal Hamming Distances

The computed distances for all pairs are:

$$\{3, 4, 4, 1, 3, 2\}$$

- Minimal Hamming distance: $l_{\text{min}} = 1$
- Maximal Hamming distance: $l_{\text{max}} = 4$

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Conclusion

For the set of bitstrings

$$S = \{001010, 110010, 110000, 010001\},$$

the **minimal Hamming distance** is:

$$\boxed{l_{\text{min}} = 1},$$

and the **maximal Hamming distance** is:

$$\boxed{l_{\text{max}} = 4}.$$